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In-Person Conference Audio Capture Device Selection

Abstract

A conference room device located in a conference room uses a conference room microphone to capture audio for one or more in-person conference participants located in the conference room during a conference. During the conference, audio is sampled from the microphone used by the conference room device and from microphones of one or more user devices determined to be located in the conference room. Based on a score calculated for the audio sampled from a user device being higher than a score calculated for audio sampled from the conference room device, the microphone of the user device is selected for audio capture for one or more of the in-person conference participants. The audio is output through conferencing software to which the conference room device is connected via a user interface tile for the conference room without displaying a new user interface tile.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 18/322,360, filed May 23, 2023, which is a continuation of U.S. patent application Ser. No. 17/588,752, filed Jan. 31, 2022, the entire disclosures of which are hereby incorporated by reference.

FIELD

[0002] This disclosure relates to audio capture device selection in a conference, such as when one or more users in a conference room have one or more associated user devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

[0004] FIG. 1 is a block diagram of an example of an electronic computing and communications system.

[0005] FIG. 2 is a block diagram of an example internal configuration of a computing device of an electronic computing and communications system.

[0006] FIG. 3 is a block diagram of an example of a software platform implemented by an electronic computing and communications system.

[0007] FIG. 4 is a block diagram of an example of a conferencing system.

[0008] FIG. 5 is a flow diagram of an example of a method for audio capture device selection for in-person conference participants.

[0009] FIG. 6 is a flow diagram of an example of a method for keyword or phrase detection for in-person conference audio capture device selection.

[0010] FIG. 7 is a flow diagram of another example of a method for keyword or phrase detection for in-person conference audio capture device selection.

[0011] FIGS. 8A-B are illustrations of examples of user interfaces (UIs) output for display for in-person conference audio capture device selection.

DETAILED DESCRIPTION

[0012] Conferencing software, such as of a software platform (e.g., a unified communications as a service (UCAas) platform), is often used to facilitate real-time communications between user devices at different locations. The conferencing software enables user devices to join and participate in a conference in-person (e.g., from a conference room with one or more other participants) or remotely (e.g., from a location that is on-premises and not within the conference room, or from a location that is at a different geographical location than the conference room).

[0013] Typically, a conference room within which in-person conference participants are located includes a device which connects to the conferencing software that implements a conference. The device in many cases includes or otherwise uses a single audio capture device, for example, an in-

room microphone or microphone array to capture conference audio for all of the in-person conference participants regardless of their locations within the conference room. When an audio signal captured within the conference room is weak, or if there is some interference that negatively affects the audio quality thereof, it affects the overall quality of the conference, especially for remote participants who cannot hear the in-person participants well.

[0014] In particular, conventional conferencing systems are not designed to improve audio quality for in-person participants. For example, in large conference rooms, a typical device connected to conferencing software is unable to adequately detect audio from participants that may be seated at a substantial distance from the in-room microphone or otherwise adjust the audio for such participants to improve the quality thereof for others. In particular, software solutions available for improving audio capture, such as by adjusting gain values of the microphone or microphone array used by the device in the conference room, may be ineffective to improve the audio quality for the in-person conference participants. While there may in some cases be other devices with their own microphones present within the conference room, conventional conferencing systems do not leverage them or their microphones.

[0015] Implementations of this disclosure address problems such as these by leveraging one or more user devices within a conference room to capture audio of one or more in-person conference participants within the conference room for output to conferencing software. For example, a conference system may include a conference room device that configures a microphone of a user device for audio capture during a conference within a conference room associated with the conference room device. The conference room device may be configured to run a client application that connects to conferencing software (e.g., at a server) to allow all of the participants physically present within the conference room to participate in the conference with one or more remote participants connecting on their own remote devices from remote locations. In many cases, one or more of those in-person participants may have with them a user device in the conference room during the conference. The system, such as via the client application running at the conference room device or otherwise, determines that the user device is within the conference room and obtains permission from the user thereof to use the microphone of the user device. The microphone of the user device may thereafter be used to capture audio for one or more of the in-person participants within the conference room. The use of the microphone of the user device by the conference room device in such a case refers to the audio captured at the user device being output to connected devices using the conferencing software as the audio for the conference room (e.g., in connection with a UI tile of the conferencing software associated with the conference room).

Alternatively, the audio captured using the microphone of the user device can be combined with audio captured using one or more other microphones to form a combined audio signal that is then output to the connected devices using the conferencing software. The implementations described herein may require authorization of an account administrator prior to use.

[0016] To describe some implementations in greater detail, reference is first made to examples of hardware and software structures used to implement audio capture device selection for in-person conference participants. FIG. 1 is a block diagram of an example of an electronic computing and communications system **100**, which can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.

[0017] The system **100** includes one or more customers, such as customers **102A** through **102B**, which may each be a public entity, private entity, or another corporate entity or individual that purchases or otherwise uses software services, such as of a UCaaS platform provider. Each customer can include one or more clients. For example, as shown and without limitation, the customer **102A** can include clients **104A** through **104B**, and the customer **102B** can include clients **104C** through **104D**. A customer can include a customer network or domain. For example, and without limitation, the clients **104A** through **104B** can be associated or communicate with a

customer network or domain for the customer **102A** and the clients **104C** through **104D** can be associated or communicate with a customer network or domain for the customer **102B**.

[0018] A client, such as one of the clients **104A** through **104D**, may be or otherwise refer to one or both of a client device or a client application. Where a client is or refers to a client device, the client can comprise a computing system, which can include one or more computing devices, such as a mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices. Where a client instead is or refers to a client application, the client can be an instance of software running on a customer device (e.g., a client device or another device). In some implementations, a client can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients.

[0019] The system **100** can include a number of customers and/or clients or can have a configuration of customers or clients different from that generally illustrated in FIG. **1**. For example, and without limitation, the system **100** can include hundreds or thousands of customers, and at least some of the customers can include or be associated with a number of clients.

[0020] The system **100** includes a datacenter **106**, which may include one or more servers. The datacenter **106** can represent a geographic location, which can include a facility, where the one or more servers are located. The system **100** can include a number of datacenters and servers or can include a configuration of datacenters and servers different from that generally illustrated in FIG. **1**. For example, and without limitation, the system **100** can include tens of datacenters, and at least some of the datacenters can include hundreds or another suitable number of servers. In some implementations, the datacenter **106** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than the customer domains for the customers **102A** through **102B**.

[0021] The datacenter **106** includes servers used for implementing software services of a UCaaS platform. The datacenter **106** as generally illustrated includes an application server **108**, a database server **110**, and a telephony server **112**. The servers **108** through **112** can each be a computing system, which can include one or more computing devices, such as a desktop computer, a server computer, or another computer capable of operating as a server, or a combination thereof. A suitable number of each of the servers **108** through **112** can be implemented at the datacenter **106**. The UCaaS platform uses a multi-tenant architecture in which installations or instantiations of the servers **108** through **112** is shared amongst the customers **102A** through **102B**.

[0022] In some implementations, one or more of the servers **108** through **112** can be a non-hardware server implemented on a physical device, such as a hardware server. In some implementations, a combination of two or more of the application server **108**, the database server **110**, and the telephony server **112** can be implemented as a single hardware server or as a single non-hardware server implemented on a single hardware server. In some implementations, the datacenter **106** can include servers other than or in addition to the servers **108** through **112**, for example, a media server, a proxy server, or a web server.

[0023] The application server **108** runs web-based software services deliverable to a client, such as one of the clients **104A** through **104D**. As described above, the software services may be of a UCaaS platform. For example, the application server **108** can implement all or a portion of a UCaaS platform, including conferencing software, messaging software, and/or other intra-party or inter-party communications software. The application server **108** may, for example, be or include a unitary Java Virtual Machine (JVM).

[0024] In some implementations, the application server **108** can include an application node, which can be a process executed on the application server **108**. For example, and without limitation, the application node can be executed in order to deliver software services to a client, such as one of the clients **104A** through **104D**, as part of a software application. The application node can be implemented using processing threads, virtual machine instantiations, or other computing features

of the application server **108**. In some such implementations, the application server **108** can include a suitable number of application nodes, depending upon a system load or other characteristics associated with the application server **108**. For example, and without limitation, the application server **108** can include two or more nodes forming a node cluster. In some such implementations, the application nodes implemented on a single application server **108** can run on different hardware servers.

[0025] The database server **110** stores, manages, or otherwise provides data for delivering software services of the application server **108** to a client, such as one of the clients **104A** through **104D**. In particular, the database server **110** may implement one or more databases, tables, or other information sources suitable for use with a software application implemented using the application server **108**. The database server **110** may include a data storage unit accessible by software executed on the application server **108**. A database implemented by the database server **110** may be a relational database management system (RDBMS), an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. The system **100** can include one or more database servers, in which each database server can include one, two, three, or another suitable number of databases configured as or comprising a suitable database type or combination thereof.

[0026] In some implementations, one or more databases, tables, other suitable information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the system **100** other than the database server **110**, for example, the client **104** or the application server **108**.

[0027] The telephony server **112** enables network-based telephony and web communications from and to clients of a customer, such as the clients **104A** through **104B** for the customer **102A** or the clients **104C** through **104D** for the customer **102B**. Some or all of the clients **104A** through **104D** may be voice over internet protocol (VOIP)-enabled devices configured to send and receive calls over a network **114**. In particular, the telephony server **112** includes a session initiation protocol (SIP) zone and a web zone. The SIP zone enables a client of a customer, such as the customer **102A** or **102B**, to send and receive calls over the network **114** using SIP requests and responses. The web zone integrates telephony data with the application server **108** to enable telephony-based traffic access to software services run by the application server **108**. Given the combined functionality of the SIP zone and the web zone, the telephony server **112** may be or include a cloud-based private branch exchange (PBX) system.

[0028] The SIP zone receives telephony traffic from a client of a customer and directs same to a destination device. The SIP zone may include one or more call switches for routing the telephony traffic. For example, to route a VOIP call from a first VOIP-enabled client of a customer to a second VOIP-enabled client of the same customer, the telephony server **112** may initiate a SIP transaction between a first client and the second client using a PBX for the customer. However, in another example, to route a VOIP call from a VOIP-enabled client of a customer to a client or non-client device (e.g., a desktop phone which is not configured for VOIP communication) which is not VOIP-enabled, the telephony server **112** may initiate a SIP transaction via a VOIP gateway that transmits the SIP signal to a public switched telephone network (PSTN) system for outbound communication to the non-VOIP-enabled client or non-client phone. Hence, the telephony server **112** may include a PSTN system and may in some cases access an external PSTN system.

[0029] The telephony server **112** includes one or more session border controllers (SBCs) for interfacing the SIP zone with one or more aspects external to the telephony server **112**. In particular, an SBC can act as an intermediary to transmit and receive SIP requests and responses between clients or non-client devices of a given customer with clients or non-client devices external to that customer. When incoming telephony traffic for delivery to a client of a customer, such as one of the clients **104A** through **104D**, originating from outside the telephony server **112** is

received, a SBC receives the traffic and forwards it to a call switch for routing to the client.

[0030] In some implementations, the telephony server **112**, via the SIP zone, may enable one or more forms of peering to a carrier or customer premise. For example, Internet peering to a customer premise may be enabled to ease the migration of the customer from a legacy provider to a service provider operating the telephony server **112**. In another example, private peering to a customer premise may be enabled to leverage a private connection terminating at one end at the telephony server **112** and at the other end at a computing aspect of the customer environment. In yet another example, carrier peering may be enabled to leverage a connection of a peered carrier to the telephony server **112**.

[0031] In some such implementations, a SBC or telephony gateway within the customer environment may operate as an intermediary between the SBC of the telephony server **112** and a PSTN for a peered carrier. When an external SBC is first registered with the telephony server **112**, a call from a client can be routed through the SBC to a load balancer of the SIP zone, which directs the traffic to a call switch of the telephony server **112**. Thereafter, the SBC may be configured to communicate directly with the call switch.

[0032] The web zone receives telephony traffic from a client of a customer, via the SIP zone, and directs same to the application server **108** via one or more Domain Name System (DNS) resolutions. For example, a first DNS within the web zone may process a request received via the SIP zone and then deliver the processed request to a web service which connects to a second DNS at or otherwise associated with the application server **108**. Once the second DNS resolves the request, it is delivered to the destination service at the application server **108**. The web zone may also include a database for authenticating access to a software application for telephony traffic processed within the SIP zone, for example, a softphone.

[0033] The clients **104A** through **104D** communicate with the servers **108** through **112** of the datacenter **106** via the network **114**. The network **114** can be or include, for example, the Internet, a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client and one or more servers. In some implementations, a client can connect to the network **114** via a communal connection point, link, or path, or using a distinct connection point, link, or path. For example, a connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

[0034] The network **114**, the datacenter **106**, or another element, or combination of elements, of the system **100** can include network hardware such as routers, switches, other network devices, or combinations thereof. For example, the datacenter **106** can include a load balancer **116** for routing traffic from the network **114** to various servers associated with the datacenter **106**. The load balancer **116** can route, or direct, computing communications traffic, such as signals or messages, to respective elements of the datacenter **106**.

[0035] For example, the load balancer **116** can operate as a proxy, or reverse proxy, for a service, such as a service provided to one or more remote clients, such as one or more of the clients **104A** through **104D**, by the application server **108**, the telephony server **112**, and/or another server. Routing functions of the load balancer **116** can be configured directly or via a DNS. The load balancer **116** can coordinate requests from remote clients and can simplify client access by masking the internal configuration of the datacenter **106** from the remote clients.

[0036] In some implementations, the load balancer **116** can operate as a firewall, allowing or preventing communications based on configuration settings. Although the load balancer **116** is depicted in FIG. **1** as being within the datacenter **106**, in some implementations, the load balancer **116** can instead be located outside of the datacenter **106**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both within and outside of the datacenter **106**. In some implementations, the load balancer **116** can be omitted.

[0037] FIG. 2 is a block diagram of an example internal configuration of a computing device **200** of an electronic computing and communications system. In one configuration, the computing device **200** may implement one or more of the client **104**, the application server **108**, the database server **110**, or the telephony server **112** of the system **100** shown in FIG. 1.

[0038] The computing device **200** includes components or units, such as a processor **202**, a memory **204**, a bus **206**, a power source **208**, peripherals **210**, a UI **212**, a network interface **214**, other suitable components, or a combination thereof. One or more of the memory **204**, the power source **208**, the peripherals **210**, the UI **212**, or the network interface **214** can communicate with the processor **202** via the bus **206**.

[0039] The processor **202** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **202** can include another type of device, or multiple devices, configured for manipulating or processing information. For example, the processor **202** can include multiple processors interconnected in one or more manners, including hardwired or networked. The operations of the processor **202** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **202** can include a cache, or cache memory, for local storage of operating data or instructions.

[0040] The memory **204** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory can be random access memory (RAM) (e.g., a DRAM module, such as DDR SDRAM). In another example, the non-volatile memory of the memory **204** can be a disk drive, a solid state drive, flash memory, or phase-change memory. In some implementations, the memory **204** can be distributed across multiple devices. For example, the memory **204** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices.

[0041] The memory **204** can include data for immediate access by the processor **202**. For example, the memory **204** can include executable instructions **216**, application data **218**, and an operating system **220**. The executable instructions **216** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be executed by the processor **202**. For example, the executable instructions **216** can include instructions for performing some or all of the techniques of this disclosure. The application data **218** can include user data, database data (e.g., database catalogs or dictionaries), or the like. In some implementations, the application data **218** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof. The operating system **220** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®; an operating system for a mobile device, such as a smartphone or tablet device; or an operating system for a non-mobile device, such as a mainframe computer.

[0042] The power source **208** provides power to the computing device **200**. For example, the power source **208** can be an interface to an external power distribution system. In another example, the power source **208** can be a battery, such as where the computing device **200** is a mobile device or is otherwise configured to operate independently of an external power distribution system. In some implementations, the computing device **200** may include or otherwise use multiple power sources. In some such implementations, the power source **208** can be a backup battery.

[0043] The peripherals **210** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **200** or the environment around the computing device **200**. For example, the peripherals **210** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **200**, such as the processor **202**. In some implementations, the computing device **200** can omit the peripherals **210**.

[0044] The UI **212** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen,

or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, or other suitable display.

[0045] The network interface **214** provides a connection or link to a network (e.g., the network **114** shown in FIG. **1**). The network interface **214** can be a wired network interface or a wireless network interface. The computing device **200** can communicate with other devices via the network interface **214** using one or more network protocols, such as using Ethernet, transmission control protocol (TCP), internet protocol (IP), power line communication, an IEEE 802.X protocol (e.g., Wi-Fi, Bluetooth, or ZigBee), infrared, visible light, general packet radio service (GPRS), global system for mobile communications (GSM), code-division multiple access (CDMA), Z-Wave, another protocol, or a combination thereof.

[0046] FIG. **3** is a block diagram of an example of a software platform **300** implemented by an electronic computing and communications system, for example, the system **100** shown in FIG. **1**. The software platform **300** is a UCaaS platform accessible by clients of a customer of a UCaaS platform provider, for example, the clients **104A** through **104B** of the customer **102A** or the clients **104C** through **104D** of the customer **102B** shown in FIG. **1**. The software platform **300** may be a multi-tenant platform instantiated using one or more servers at one or more datacenters including, for example, the application server **108**, the database server **110**, and the telephony server **112** of the datacenter **106** shown in FIG. **1**.

[0047] The software platform **300** includes software services accessible using one or more clients. For example, a customer **302** as shown includes four clients—a desk phone **304**, a computer **306**, a mobile device **308**, and a shared device **310**. The desk phone **304** is a desktop unit configured to at least send and receive calls and includes an input device for receiving a telephone number or extension to dial to and an output device for outputting audio and/or video for a call in progress. The computer **306** is a desktop, laptop, or tablet computer including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The mobile device **308** is a smartphone, wearable device, or other mobile computing aspect including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The desk phone **304**, the computer **306**, and the mobile device **308** may generally be considered personal devices configured for use by a single user. The shared device **310** is a desk phone, a computer, a mobile device, or a different device which may instead be configured for use by multiple specified or unspecified users.

[0048] Each of the clients **304** through **310** includes or runs on a computing device configured to access at least a portion of the software platform **300**. In some implementations, the customer **302** may include additional clients not shown. For example, the customer **302** may include multiple clients of one or more client types (e.g., multiple desk phones or multiple computers) and/or one or more clients of a client type not shown in FIG. **3** (e.g., wearable devices or televisions other than as shared devices). For example, the customer **302** may have tens or hundreds of desk phones, computers, mobile devices, and/or shared devices.

[0049] The software services of the software platform **300** generally relate to communications tools, but are in no way limited in scope. As shown, the software services of the software platform **300** include telephony software **312**, conferencing software **314**, messaging software **316**, and other software **318**. Some or all of the software **312** through **318** uses customer configurations **320** specific to the customer **302**. The customer configurations **320** may, for example, be data stored within a database or other data store at a database server, such as the database server **110** shown in FIG. **1**.

[0050] The telephony software **312** enables telephony traffic between ones of the clients **304** through **310** and other telephony-enabled devices, which may be other ones of the clients **304** through **310**, other VOIP-enabled clients of the customer **302**, non-VOIP-enabled devices of the customer **302**, VOIP-enabled clients of another customer, non-VOIP-enabled devices of another

customer, or other VOIP-enabled clients or non-VOIP-enabled devices. Calls sent or received using the telephony software **312** may, for example, be sent or received using the desk phone **304**, a softphone running on the computer **306**, a mobile application running on the mobile device **308**, or using the shared device **310** that includes telephony features.

[0051] The telephony software **312** further enables phones that do not include a client application to connect to other software services of the software platform **300**. For example, the telephony software **312** may receive and process calls from phones not associated with the customer **302** to route that telephony traffic to one or more of the conferencing software **314**, the messaging software **316**, or the other software **318**.

[0052] The conferencing software **314** enables audio, video, and/or other forms of conferences between multiple participants, such as to facilitate a conference between those participants. In some cases, the participants may all be physically present within a single location, for example, a conference room, in which the conferencing software **314** may facilitate a conference between only those participants and using one or more clients within the conference room. In some cases, one or more participants may be physically present within a single location and one or more other participants may be remote, in which the conferencing software **314** may facilitate a conference between all of those participants using one or more clients within the conference room and one or more remote clients. In some cases, the participants may all be remote, in which the conferencing software **314** may facilitate a conference between the participants using different clients for the participants. The conferencing software **314** can include functionality for hosting, presenting scheduling, joining, or otherwise participating in a conference. The conferencing software **314** may further include functionality for recording some or all of a conference and/or documenting a transcript for the conference.

[0053] The messaging software **316** enables instant messaging, unified messaging, and other types of messaging communications between multiple devices, such as to facilitate a chat or other virtual conversation between users of those devices. The unified messaging functionality of the messaging software **316** may, for example, refer to email messaging which includes a voicemail transcription service delivered in email format.

[0054] The other software **318** enables other functionality of the software platform **300**. Examples of the other software **318** include, but are not limited to, device management software, resource provisioning and deployment software, administrative software, third party integration software, and the like. In one particular example, the other software **318** can enable audio capture device selection for in-person conference participants. In such a case, the other software **318** may optionally be included in the conferencing software **314**.

[0055] The software **312** through **318** may be implemented using one or more servers, for example, of a datacenter such as the datacenter **106** shown in FIG. **1**. For example, one or more of the software **312** through **318** may be implemented using an application server, a database server, and/or a telephony server, such as the servers **108** through **112** shown in FIG. **1**. In another example, one or more of the software **312** through **318** may be implemented using servers not shown in FIG. **1**, for example, a meeting server, a web server, or another server. In yet another example, one or more of the software **312** through **318** may be implemented using one or more of the servers **108** through **112** and one or more other servers. The software **312** through **318** may be implemented by different servers or by the same server.

[0056] Features of the software services of the software platform **300** may be integrated with one another to provide a unified experience for users. For example, the messaging software **316** may include a UI element configured to initiate a call with another user of the customer **302**. In another example, the telephony software **312** may include functionality for elevating a telephone call to a conference. In yet another example, the conferencing software **314** may include functionality for sending and receiving instant messages between participants and/or other users of the customer **302**. In yet another example, the conferencing software **314** may include functionality for file

sharing between participants and/or other users of the customer **302**. In some implementations, some or all of the software **312** through **318** may be combined into a single software application run on clients of the customer, such as one or more of the clients **304** through **310**.

[0057] FIG. **4** is a block diagram of an example of a conferencing system **400**. The conferencing system **400** includes a server **402**. The server **402** runs conferencing software, for example, the conferencing software **314** shown in FIG. **3**. For example, the server **402** may be the application server **108** shown in FIG. **1**. As shown, a conference implemented using the server **402** includes in-person participants located within a conference room **404** and remote participants connected via remote user devices **1 406A** through **N** (in which **N** is an integer greater than or equal to **2**) **406N**. The conferencing system **400** is configured to detect when an audio signal captured using a conference room microphone **408** located within the conference room **404** is weak and/or negatively affected by some interference. Based on such detection, the conferencing system configures one or more user devices in the conference room **404** to capture audio to improve the audio quality of the conference. The conferencing system **400** is configured to output the audio captured at the user device within the conferencing software running at the server **402** as the audio for the conference room **404**.

[0058] The server **402** is configured to communicate with the conference room **404** via a conference room device **410**. The conference room device **410** is a computing device configured for the conference room **404**. The conference room device **410** may generally be a computer such as a laptop or desktop computer located within the conference room **404**, and in some cases, may be a mobile device configured for conferencing purposes. In some cases, the conference room device **410** may be located outside of the conference room **404**. The conference room device **410** runs a client application that connects to conferencing software running at the server **402** to enable the in-person participants physically present within the conference room **404** to participate in a conference with one or more remote participants connecting on the remote user devices **1 406A** through **N 406N**. The conference room microphone **408** is or includes a microphone or microphone array. The conference room microphone **408** may be integrated within the conference room device. Alternatively, the conference room microphone **408** may be a separate device or other component which is coupled to the conference room device using a wired or wireless connection.

[0059] As shown in FIG. **4**, three in-person participants, User **1**, User **2**, and User **N** are located in the conference room **404**. The conference room **404** can contain any number of participants based on the size of the conference room **404**, and three participants are simply shown for clarity. Each participant may be associated with one or more user devices. In some cases, a participant may not have an associated user device. In this example, User **1** may be associated with user device **1 (UD1) 412A** and user device **2 (UD2) 412B**, User **3** may be associated with user device **N (UDN) 412N**, and User **2** may not be associated with a user device. Associations between user devices and respective participants can be determined, for example, based on user accounts of those participants registered with the conferencing software run on the server **402** or a software platform which implements the conferencing software. For example, information stored in connection with such a user account may identify devices which have been logged into by the subject user. Where such a device is present in the conference room **404**, it may be considered an associated user device.

[0060] In an example, the quality of audio captured for User **1** using the conference room microphone **408** may be poor because User **1** is located at a distance from the conference room microphone **408** (e.g., outside a range of the conference room microphone **408** defined for quality audio capture). In this example, the conference room device **410** or the server **402** may determine that the audio quality captured for User **1** is below a threshold by sampling audio signals using the conference room microphone **408**. The audio quality captured for one or more of the User **1**, User **2**, or User **N** may be determined by calculating a score, for example, a voice quality score (VQS), of the audio captured by the conference room microphone **408**. A VQS is a score calculated to represent the perceptible quality of a voice of one or more participants. The VQS may range from a

score of 1.0 to 5.0 and may be based on a mean opinion score (MOS) of the audio captured by the conference room microphone **408**, where a score of 1.0 represents a scenario where the audio quality makes it impossible to communicate and a score of 5.0 represents a scenario where the audio quality is similar to a face-to-face conversation. In some examples, the VQS may be based on a signal-to-noise ratio (SNR) of the audio captured by the conference room microphone **408**. In some examples, the VQS may be based on another VQS of the audio captured by the conference room microphone **408**. In some examples, the VQS may be based on any combination of the MOS, SNR, and the other VQS. In an example, the threshold may be a baseline VQS score of 3.5.

[0061] At some point in time, either prior to the conference or during the conference, the conference room device **410** or the server **402** may transmit a prompt to a user device associated with an in-person conference participant. The prompt may be a visual notification, and audio notification, a haptic notification, or any combination thereof. The prompt may be a request for permission to access a microphone of the user device, a request for the participant to place the user device on a conference room table, or both. For example, a prompt may be transmitted to the UD N **412N** of the User N. In another example, a prompt may be transmitted to one or both of the UD 1 **412A** or the UD 2 **412B** of the User 1. The prompt used to request permission to access the microphone of a given user device is based on the conference and may be generally understood to be automatically revoked at an end of the conference. As such, the prompt is not a prompt which may be presented in connection with the download or installation of a software application (e.g., a client application) on the user device, such as to configure the general use of the software application. The automatic revocation of permission to access the microphone of a given user device may be determined based on a scheduled end time for the conference (e.g., according to scheduling/calendar data), a time at which the conference is actually ended using the conferencing software, or upon some duration of time (e.g., five minutes) passing without audio being captured after audio capture has occurred for the conference. The revocation of permission to access the microphone of a given user device may be subject to a user prompt, for example, the user may be requested to click a button if the conference is ongoing, in which case if the conference is still ongoing based on a user input, an expiration of the duration of time will not close the conference.

[0062] When permission to access the microphone of a given user device is received, the conference room device **410** or the server **402** may sample audio using the microphone of the user device. The conference room device **410** or the server **402** is configured to determine a VQS for the audio sampled using the microphone of the user device. This process is performed for each user device for which permission to access the microphone is granted by the user thereof.

[0063] The conference room device **410** or the server **402** are configured to determine whether the VQS of the audio from a given user device located within the conference room **404** is greater than the VQS of the audio from the conference room microphone **408**. Based on a determination that the VQS of the audio from a user device is greater than the VQS of the audio from the conference room microphone **408**, the conference room device **410** or the server **402** causes the microphone of the user device to capture audio for output to the conference.

[0064] In some implementations, the conference room device **410** or the server **402** may sample audio using the microphones of some or all of the user devices located within the conference room **404** for individual in-person participants, for example, based on those participants being more than a threshold distance away from the conference room microphone **408**. In some implementations, the conference room device **410** or the server **402** may sample audio using the microphones of some or all the user devices located within the conference room **404** for all of the in-person participants. The sampling of the audio using the microphones of the user devices may be performed sequentially by user device or for all user devices substantially simultaneously. The amount of audio sampled from a given user device may vary.

[0065] In conventional conferencing software implementations, a different UI tile is added to the conferencing software UI for each device from which media of the conference is received. By way

of contrast, in the implementations disclosed herein, the in-person participants may be displayed in a conferencing software UI as a single UI tile for the conference room **404**. As such, the use of a microphone other than the conference room microphone **408** will not cause a new UI tile to be displayed for the conference room **404**, rather the audio captured using the other microphone is associated with the existing conference room UI tile.

[0066] In many cases, one or more of the in-person participants may have with them one or more user devices in the conference room **404** during the conference that is idle. The conference room device **410** or the server **402** configures a microphone of one or more of the user devices within the conference room **404** to capture audio for one or more of the in-person participants within the conference room.

[0067] In an example use case, User **1** may be located at a distance outside the range of the conference room microphone **408**. As shown in FIG. **4**, User **1** has two associated user devices, UD1 **412A** and UD2 **412B**. In this example, the conference room device **410** or the server **402** may determine that the audio quality captured for User **1** is below a threshold by sampling audio signals using the conference room microphone **408** and determining a VQS for the audio.

[0068] At some point in time, either prior to the conference or during the conference, the User **1** may authorize the conference room device **410** to access a microphone of the UD1 **412A**, the UD2 **412B**, or both. The conference room device **410** or the server **402** may sample audio using the conference room microphone **408** and the microphone of the UD1 **412A**, the UD2 **412B**, or both. The conference room device **410** or the server **402** are configured to determine a VQS for the audio from User **1** based on the sampled audio from the microphone of the UD1 **412A**, the microphone of the UD2 **412B**, or both.

[0069] The conference room device **410** or the server **402** are configured to determine whether the VQS of the audio from the UD1 **412A**, the VQS of the audio from the UD2 **412B**, or both, are greater than the VQS of the conference room microphone **408**. In this example, the conference room device **410** or the server **402** may determine that the audio from the UD2 **412B** has the highest VQS. Based on a determination that the VQS of the audio from the UD2 **412B** is greater than the VQS of the audio from the conference room microphone **408**, the conference room device **410** or the server **402** are configured to capture audio for User **1** using the microphone of the UD2 **412B** for output to the conference software without adding a UI tile to the conference UI.

[0070] In another example use case, the audio quality captured for User **2**, who as shown does not have an associated user device, as captured using the conference room microphone **408** may be poor, for example, because User **2** is sitting outside of a preferred range of the conference room microphone **408** or because User **2** is speaking at a low volume. In this example, the conference room device **410** or the server **402** may determine that the audio quality captured for User **2** is below a threshold by sampling audio signals using the conference room microphone **408** and determining a VQS for the audio.

[0071] The conference room device **410** or the server **402** may sample audio using the conference room microphone **408** and the microphone of any user devices for which permission to access the microphone thereof has been granted. Optionally, the conference room device **410** or the server **402** may determine that the User **2** does not have an associated device.

[0072] In this example use case, the conference room device **410** or the server **402** may sample audio from User **2** using the microphone of the UD1 **412A**, the microphone of the UD2 **412B**, and the microphone of the UDN **412N**. The conference room device **410** or the server **402** are configured to determine a VQS for the audio from User **2** based on the sampled audio from the microphone of the UD1 **412A**, the microphone of the UD2 **412B**, and the microphone of the UDN **412N**.

[0073] The conference room device **410** or the server **402** are configured to determine whether the VQS of the audio from the UD1 **412A**, the VQS of the audio from the UD2 **412B**, or the VQS of the audio from the UDN **412N**, are greater than the VQS of the conference room microphone **408**.

In this example, the conference room device **410** or the server **402** may determine that the audio from the UDN **412N** has the highest VQS. Based on a determination that the VQS of the audio from the UDN **412N** is greater than the VQS of the audio from the conference room microphone **408**, the conference room device **410** or the server **402** are configured to capture audio of User **2** using the microphone of the UDN **412N** for output to the conference without adding a UI tile to the conference UI.

[0074] Alternatively, the conference room device **410** or the server **402** may determine that the VQS of the audio from the UD2 **412B** and the VQS of the audio from the UDN **412N** are greater than the VQS of the audio from the conference room microphone **408**. Based on the determination that the VQS of the audio from the UD2 **412B** and the VQS of the audio from the UDN **412N** are greater than the VQS of the audio from the conference room microphone **408**, the conference room device **410** or the server **402** are configured to use the microphone of the UD2 **412B** and the microphone of the UDN **412N**, to capture audio of User **2**, and combine the audio signals from the microphone of the UD2 **412B** and the microphone of the UDN **412N** for output to the conference without adding a UI tile to the conference UI.

[0075] As has been described, the processing herein including to prompt user devices for microphone access permission, sampling audio of user devices, determining VQSs for audio sampled from user devices, and selecting audio capture devices based on the VQSs may be performed wholly by the conference room device, wholly by the server **402**, or partially by the conference room device and partially by the server **402**. Where the processing is performed wholly by the conference room device **410**, a connection between the conference room device **410** and the UD1 **412A**, the UD2 **412B**, and the UDN **412N** may be established upon receiving permission from the respective user devices to access the microphones of the respective user devices. Once the connection between the conference room device **410** and the respective user device is established, audio may be captured from the microphone of the respective user device and transmitted to the conference room device **410**. The conference room device **410** is configured to output the audio captured using the microphone of the respective user device to the conference. Where the processing is performed wholly by the server **402**, a connection between the server **402** and the UD1 **412A**, the UD2 **412B**, and the UDN **412N** may be established upon receiving permission from the respective user devices to access the microphones of the respective user devices. Once the connection between the respective user device and the server **402** is established, the respective user device captures audio using its microphone and transmits the audio to the server **402**. The server **402** is configured to output the audio captured using the microphone of the respective user device to the conference. Where the processing is performed partially by each of the conference room device and the server **402**, a connection between the conference room device **410** and the UD1 **412A**, the UD2 **412B**, and the UDN **412N** may be established upon receiving permission from the respective user devices to access the microphones of the respective user devices. Once the connection between the conference room device **410** and the respective user device is established, audio may be captured from the microphone of the respective user device and transmitted to the conference room device **410**. The conference room device **410** transmits the audio captured using the microphone of the respective user device to the server **402**. The server **402** outputs the audio captured using the microphone of the respective user device to the conference.

[0076] Although the above discussion is generally related to participants within a conference room, the conferencing system **400** of FIG. **4** can be used in other situations. For example, the conferencing system **400** can be used in academic settings to support audio capture for teachers and/or students within a classroom, such as to improve audio quality of the captured audio for other teachers, students, or guests virtually connected to a class session over the conferencing software. In another example, the conferencing system **400** can be used in a non-conference room office setting where there are multiple participants in one or more of those rooms. In yet another example, the conferencing system **400** can be used for expos, webinars, and other event presentations with

virtual components in which audience audio capture is important.

[0077] To further describe some implementations in greater detail, reference is next made to examples of methods that may be performed by or using a system for audio capture device selection in a conference for improved audio quality. FIGS. 5-7 are flowcharts of examples of methods for audio capture device selection for in-person conference participants. The methods **500**, **600**, and **700** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. 1-4. The methods **500**, **600**, and **700** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the methods **500**, **600**, **700**, or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

[0078] For simplicity of explanation, the methods **500**, **600**, and **700** are depicted and described herein as a series of steps or operations. However, the steps or operations in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

[0079] FIG. 5 is a flow diagram of an example of a method **500** for audio capture device selection in a conference. At **502**, the method **500** includes capturing first audio during a conference using a microphone of a conference room device located within a conference room. The first audio is audio from one or more in-person participants within the conference room that is captured by the conference room microphone. The first audio captured by the conference room microphone is output to the conference such that remote participants can listen to the audio component of the conference using their respective remote user devices.

[0080] At **504**, the method **500** includes determining whether a user device is located within the conference room. The determination that the user device is located within the conference room may be based on a number of factors including calendar information, location information (e.g., geolocation data) from the user device, detection of movement of the user device, ultrasound signal detection, or any combination thereof. The calendar information may be stored in a calendar database for storing calendar information of users of the conferencing system. The calendar information may include an invite that indicates when a user is scheduled to attend a conference in person in a conference room including start and/or end times of the conference, a location of the conference (i.e., the conference room), an invite status (i.e., an indication of whether the user has accepted the invite), or any combination thereof. At the start time of the conference, or at a predetermined time prior to the start of the conference, the conferencing system may query the calendar database to obtain the calendar information of the in-person participants of the conference. Based on the invite status, the conferencing system may determine that the user device is located within the conference room. For example, when the invite status indicates that the invite was accepted by the user, the conferencing system determines that the user device is located within the conference room at or near the start time of the conference.

[0081] The location information may include geolocation data that is obtained from a sensor of the user device, such as a GPS sensor. The location information may show that the user is at the premises of the conference room location or in the conference room at or near the conference start time. The location information may be determined based on a beacon or some other radio signal that is transmitted by a conference room device. In an example, the user device may receive the beacon or other radio signal from the conference room device and transmit a message to the server that the beacon was received. The server may determine that the user device is located within the conference room based on the reception of the message. In an example, the calendar information, either alone or in conjunction with the location information, may be used to determine that the user device is located within the conference room. In an example where the calendar information is used

in conjunction with the location information, the conferencing system may determine that the user device is located within the conference room based on a determination that the invite status indicates that the invite was accepted and that the location information indicates that the user device is near the conference room at the start time of the conference.

[0082] The determination that the user device is located within the conference room may be based on a determination that the user device has moved within a predetermined time of a start time of the conference. The conferencing system may detect that the user device has moved by obtaining sensor information, such as accelerometer/gyroscope information, from the user device. In an example, the user device may transmit a message to the server indicating movement of the device based on the sensor information when the calendar information indicates that the invite for the conference was accepted. The determination that the user device is located within the conference room may also be based on the user device connecting to the conference (e.g., via a client on the user device).

[0083] The determination that the user device is located within the conference room may be based on a detection of an ultrasound signal transmitted by a conference room device. The user device may detect the ultrasound signal and transmit a message to the conference room device indicating that the user device detected the ultrasound signal. The conference room device may determine that the user device is present within the conference room based on the message. The conference room device may transmit a message to the server to notify the server that the user device is present within the conference room.

[0084] At **506**, the method **500** includes determining that permission to access a microphone of a user device located in the conference room for the conference has been granted. The determination that permission to access the microphone of the user device located in the conference room has been granted can occur before or after the capturing of the first audio at **502**.

[0085] In some examples, the method **500** may include transmitting a prompt to the user device at the start of the conference or during the conference based on a determination that the user device is located within the conference room. The prompt may be a visual notification, an audible notification, a haptic notification, or any combination thereof. The prompt may indicate that the audio of the conference may be improved if the user places their user device (e.g., mobile phone, tablet, or laptop) on a table within the conference room. The prompt may include a request for permission for the conference system to access the microphone of the user device to be used by the conference system to capture audio for output to the conference. When the conference system receives permission from the user device, the conference system opens an audio channel between a server of the conference system and the user device. In some examples, the conference system may open an audio channel between the conference room device and the user device. The prompt may be transmitted to the user device responsive to the determination that the user device is located within the conference room (e.g., based on one or more of the factors described above).

Alternatively, the prompt may be transmitted to the user device at some time after the determination is made that the user device is located within the conference room. For example, the prompt may be transmitted to the user device responsive to a VQS for audio captured using a conference room microphone being below a threshold. The determination that permission to access the microphone of the user device has been granted is based on receiving a message from the user device in response to the request. The message may indicate that access to the microphone of the user device has been granted. If access to the microphone of the user device is not granted, permission from other user devices may be requested or the conference may continue using the audio captured using the conference room microphone.

[0086] At **508**, the method **500** includes sampling audio signals from the microphone used by the conference room device and the microphone of the user device. The audio signals from the microphone of the conference room device, the microphone of the user device, or both, may be sampled periodically or based on a triggering event, such as detecting a keyword or phrase using

speech detection. For example, if a keyword or phrase such as “Todd, I cannot hear you,” or “Can you speak up Todd?” is detected, it may be determined that the audio capture device (e.g., the conference room device) currently used to capture audio from the conference participant named Todd may need to be switched to another audio capture device (e.g. the user device) that belongs to the conference participant or is within a vicinity of the conference participant such that it provides improved audio performance over the current audio capture device. In another example, the audio signals are sampled periodically to determine whether the quality of the audio from the microphone of the conference room device, the microphone of the user device, or both, is below a threshold. If it is determined that the audio quality from the currently used microphone is below a threshold, the conferencing system may determine that a switch to another microphone is needed.

[0087] At **510**, the method **500** includes determining a first VQS for the audio from the microphone of the conference room device and a second VQS for the audio from the microphone of the user device. In some examples, the sampling of the audio signals from the conference room device may be triggered when a VQS of the audio from the microphone of the conference device falls below a threshold. The first VQS and the second VQS can be used to determine which microphone is the best microphone to capture audio from, for example, based on an MOS, an SNR, another VQS, or any combination thereof.

[0088] At **512**, the method **500** includes capturing second audio using the microphone of the user device based on the first VQS of the audio from the microphone of the conference room device being less than the second VQS of the audio from the microphone of the user device. The second audio is audio from one or more in-person participants within the conference room that is captured by the microphone of the user device. The second audio captured by the microphone of the user device is output to the conference such that remote participants can listen to the audio component of the conference using their respective remote user devices. Capturing the second audio using the microphone of the user device may thus be understood to include determining to capture the second audio using the microphone of the user device. In some examples, the capturing of the second audio using the microphone of the user device may be based on a difference between the first VQS and the second VQS being above a threshold. In at least some cases, based on the determination to capture the second audio using the microphone of the user device, a brief training for echo cancellation may be performed prior to actually capturing the second audio using the microphone of the user device.

[0089] An echo canceller of the system learns the direction and latency of the echo produced when feedback from the cross chatter from multiple microphones causes an echo and diminishes audio quality. The echo canceller is configured to determine a distance between the user and the microphone of the conference room and a distance between the user and the microphone of the user device. Typically, the distance between the user and the microphone of the conference room is greater than the distance between the user and the microphone of the user device. The echo canceller is configured to determine, based on the learned direction and latency (e.g., caused by the determined distances) of various microphone signals, which of the microphone signal levels can be boosted or suppressed (e.g., by using weights) to cancel out the echo. The echo path (i.e., distance) can change materially when a switch to the microphone of the user device is made, therefore the echo canceller is configured to retrain to reduce echo based on the distance between the microphone of the user device and the user prior to switching to the microphone of the user device.

[0090] During the retraining, the audio captured from the microphone of the conference room is output to the conference as the system prepares to transition to capturing the audio from the microphone of the user device. During this transition process, the system is configured to open a channel between the user device and the server, perform the echo cancellation process to cancel out the echo, and switch to capturing audio from the microphone of the user device when a desirable level of echo cancellation is achieved or when the echo is stabilized.

[0091] When the second audio is being captured using the microphone of the user device, the audio

signals from the user device are transmitted to the server of the conference system or the conference room device via the audio channel opened between the user device and the server or conference room device. The server then transmits the audio captured by the microphone of the user device for output to the conference without adding a UI tile to the conference UI.

[0092] In some implementations, audio captured from microphones of multiple user devices may be sampled. The sampling of the audio captured from microphones of multiple user devices may be performed sequentially or substantially simultaneously. For example, if it is determined that the audio quality from the microphone of the conference room device is below a threshold, the conferencing system may sample a microphone for a user device to determine whether the audio quality from the microphone of the user device is greater than the audio quality from the microphone of the conference room device. If it is determined that the audio quality from the microphone of the user device is not greater than the audio quality from the microphone of the conference room device, the conferencing system may sample audio captured from a microphone of another user device and repeat the process until the audio quality from a microphone of a user device is found to be greater than the audio quality from the microphone of the conference room device. Alternatively, the conferencing system may sample audio from the microphones of all the user devices determined to be within the conference room simultaneously or substantially simultaneously.

[0093] FIG. 6 is a flow diagram of an example of a method **600** for keyword or phrase detection in a conference for audio capture device selection. In some cases, the method **600** may be implemented in conjunction with the method **500** shown in FIG. 5, for example, at operation **502**. At **602**, the method **600** includes detecting a keyword or phrase in audio of a conference. A component of the conference includes a conference room that contains one or more users (e.g., in-person conference participants) attending in person and located within the conference room. The audio that is output to the conference may be obtained using one or more microphones of a user device located within the conference room or a microphone of a conference room device. The keyword or phrase may be detected using speech detection.

[0094] The keyword or phrase is a word or group of words detected in an audio stream of the conference that indicates that the audio quality for one or more in-person conference participants is low. The conferencing system may perform voice detection to detect whether the audio stream includes speech. If it is determined that the audio stream includes speech, the conferencing system performs a speech-to-text conversion and determines whether the text includes a keyword or phrase by comparing the text to a database that stores keywords and phrases. The conferencing system may process the text using a machine learning (ML) algorithm to determine new keywords and phrases and update the database accordingly. In an example, the keyword or phrase may include an identifier associated with an in-person conference participant and an indication that the audio quality captured for the in-person conference participant is low. For example, in the phrase “Your voice sounds distant, Steve,” which may be spoken by a remote conference participant (i.e., someone who is not located in the conference room with Steve), the identifier associated with the in-person conference participant may be the name of the in-person conference participant (i.e., “Steve”), and the indication that the audio quality is low may be the portion of the phrase “Your voice sounds distant.”

[0095] At **604**, the method **600** includes identifying the in-person conference participant. Identifying the in-person conference participant may include parsing the detected phrase to determine an identifier associated with an in-person conference participant. In this example, the in-person conference participant may be identified by detecting the name of the in-person conference participant in the phrase. The identification of the in-person conference participant may be based on a determination of whether the in-person conference participant is located within the conference room. For example, if the identifier associated with the in-person conference participant is “Steve,” the conferencing system will search for an in-person conference participant named Steve that is

located within the conference room, for example, based on a prior determination of the in-person conference participants. If there are multiple in-person conference participants associated with the identifier located within the conference room, the conferencing system may identify each of these in-person conference participants as a candidate in-person conference participant. In some examples, if there are multiple in-person conference participants associated with the identifier located within the conference room, the conferencing system may sample audio from each user device associated with each of the candidate in-person conference participants that has granted access to the microphones of their respective user devices to identify the in-person conference participant.

[0096] At **606**, the method **600** includes identifying a user device associated with the identified in-person conference participant. The user device may be identified based on a user account, device location, or both. For example, the user device may be identified by determining that the user device is associated with a user account of a user that is determined to be within the conference room, for example, based on calendar information, location information, and/or device movement information. In some examples, more than one device (e.g., a mobile phone and a laptop) may be identified for a given user.

[0097] At **608**, the method **600** includes sampling audio using a microphone of the identified user device. The audio may be sampled from the start of the audio capture and for a predetermined time after speech is detected. In some cases, where applicable, multiple microphones of the user device may be used to sample the audio.

[0098] At **610**, the method **600** includes determining a VQS for the audio from the microphone of the identified user device. In some cases where the identified user device has multiple microphones, a VQS for the audio from each microphone may be determined. A determination may then be made as to which microphone or combination of microphones produces the best audio signal based on their respective VQSs.

[0099] At **612**, the method **600** includes determining whether the VQS of the audio from the microphone(s) of the user device is greater than a determined VQS of the audio from the microphone of the conference room device. If the VQS of the audio from the microphone(s) of the user device is determined to be greater than the VQS of the audio from the microphone of the conference room device, then the method **600** includes determining to capture audio using a microphone(s) of the user device and capturing audio using the microphone(s) of the user device for output to the conference at **614**. If the VQS of the audio from the microphone(s) of the user device is determined to be less than or equal to the VQS of the audio from the microphone of the conference room device, then the method **600** includes sampling audio from another user device that is located within the conference room at **616**. In some examples, the sampling of the audio may include sampling audio from all the user devices determined to be located within the conference room simultaneously or substantially simultaneously.

[0100] At **618**, the method **600** includes determining a VQS for the audio from the other user device. If the other user device has multiple microphones, a VQS for the audio from each microphone may be determined, and a determination may be made as to which microphone or combination of microphones produces the best audio signal based on their respective VQSs. In an example where audio from all the user devices determined to be within the conference room is sampled, a VQS for the audio from each of the user devices is determined.

[0101] At **620**, the method **600** includes determining whether the VQS of the audio from the other device is greater than the determined VQS of the audio from the conference room device. If the VQS of the audio from the microphone of the other user device is determined to be greater than the VQS of the audio from the microphone of the conference room device, then the method **600** includes capturing audio from the microphone of the other user device for output to the conference at **622**. If the VQS of the audio from the microphone of the other user device is determined to be less than or equal to the VQS of the audio from the microphone of the conference room device,

then the method **600** may include sampling audio from another user device at **616**. Operations **616-620** may be repeated until a suitable user device is found to capture audio for output to the conference. In an example where audio from all the user devices determined to be within the conference room is sampled, if the audio from the microphone of the user device determined to have the highest VQS is greater than the audio from the microphone of the conference room device, then the method **600** includes capturing audio using the microphone of the user device determined to have the highest VQS for output to the conference at **622**.

[0102] In some examples, it may be determined that capturing audio from the identified user device and the other user device may produce the best audio signal. In these examples, the audio signals from the microphone of the identified user device and the other user device may be combined for output to the conference. In an example, the conference room includes a long table that has a microphone of the conference room at the front of the table. The microphone of the conference room may be a speaker and microphone bar combination. In this example, a first user may be seated at or near the front of the table, and a second user may be seated at a far end of the table. The first user and the second user may be engaged in a discussion during the conference. In order to capture the audio from the second user, it may be best to capture the audio using the microphone of a user device of the second user. In this situation, however, a user device, such as a mobile phone, may not be capable of capturing audio from a user (i.e., the first user) that is at a distance from the user device of the second user. The conferencing system may enter a back-and-forth mode in which one participant's audio capture (e.g., the first user's audio capture) is through the speaker and microphone bar combination and the other participant's audio capture (e.g., the second user's audio capture) is through the user device of the second user. In order to avoid this back-and-forth mode, the conferencing system may determine the echo paths between each user and both microphones. The conferencing system may then capture audio using both microphones by combining the signals from each based on the determined echo paths to reduce echo.

[0103] FIG. 7 is a flow diagram of another example of a method **700** for keyword or phrase detection in a conference for audio capture device selection. In some cases, the method **700** may be implemented in conjunction with the method **500** shown in FIG. 5, for example, at operation **502**. At **702**, the method **700** includes detecting a keyword or phrase in audio of a conference. A component of the conference includes a conference room that contains one or more users (e.g., in-person conference participants) attending in person and located within the conference room. The audio that is output to the conference may be obtained using one or more microphones of a user device located within the conference room or a microphone of a conference room device. The keyword or phrase may be detected using speech detection.

[0104] The conferencing system may perform voice detection to detect whether the audio stream includes speech. If it is determined that the audio stream includes speech, the conferencing system performs a speech-to-text conversion and determines whether the text includes a keyword or phrase by comparing the text to a database that stores keywords or phrases. The conferencing system may process the text using an ML algorithm to determine new keywords and phrases and update the database accordingly. In an example, the keyword or phrase may include an identifier associated with an in-person conference participant and an indication that the audio quality captured for the in-person conference participant is low. For example, in the phrase “Mary, we are having trouble hearing you,” which may be spoken by a remote conference participant (i.e., someone who is not located in the conference room with Steve), the identifier associated with the in-person conference participant may be the name of the in-person conference participant (i.e., “Mary”), and the indication that the audio quality is low may be the portion of the phrase “trouble hearing you.”

[0105] At **704**, the method **700** includes identifying the in-person conference participant. Identifying the in-person conference participant may include parsing the detected phrase to determine an identifier associated with an in-person conference participant. In this example, the in-person conference participant may be identified by detecting the name of the in-person conference

participant in the phrase. The identification of the in-person conference participant may be based on a determination of whether the in-person conference participant is located within the conference room. For example, if the identifier associated with the in-person conference participant is “Mary,” the conferencing system will search for an in-person conference participant named Mary that is located within the conference room, for example, based on a prior determination of the in-person conference participants. If there are multiple in-person conference participants associated with the identifier located within the conference room, the conferencing system may identify the each of these in-person conference participants as a candidate in-person conference participant. In some examples, if there are multiple in-person conference participants associated with the identifier located within the conference room, the conferencing system may sample audio from each user device associated with each of the candidate in-person conference participants that has granted access to the microphones of their respective user devices to identify the in-person conference participant.

[0106] At **706**, the method **700** includes determining whether the in-person conference participant has an associated user device located within the conference room. This determination may be based on an identification of a user account, device location, or both. For example, it may be determined that a user device is associated with a user account of a user that is determined to be within the conference room, for example, based on calendar information, location information, and/or device movement information. In some examples, it may be determined that more than one device (e.g., a mobile phone and a laptop) is identified for a given user and is located within the conference room.

[0107] If it is determined at **706** that the identified in-person conference participant has a user device located within the conference room, the method **700** includes sampling audio using a microphone of the identified user device at **708**. The audio may be sampled from the start of the audio capture and for a predetermined time after speech is detected. In some cases, where applicable, multiple microphones of the user device may be used to sample the audio.

[0108] At **710**, the method **700** includes determining a VQS for the audio from the microphone of the identified user device. In some cases where the identified user device has multiple microphones, a VQS for the audio from each microphone may be determined. A determination may then be made as to which microphone or combination of microphones produces the best audio signal based on their respective VQSs.

[0109] At **712**, the method **700** includes determining whether the VQS of the audio from the microphone of the user device is greater than a determined VQS of the audio from the microphone of the conference room device. If the VQS of the audio from the microphone of the user device is determined to be greater than the VQS of the audio from the microphone of the conference room device, then the method **700** includes determining to capture audio using a microphone of the user device and capturing audio using the microphone of the user device for output to the conference at **714**. If the VQS of the audio from the microphone of the user device is determined to be less than or equal to the VQS of the audio from the microphone of the conference room device, then the method **700** includes sampling audio from other user devices that are located within the conference room at **716**. The other user devices may be associated with other users that are present in the conference room. In some examples, sampling audio from other user devices that are located within the conference room includes sampling all the user devices that are determined to be located within the conference room.

[0110] At **718**, the method **700** includes determining a VQS for the audio from the other user devices located within the conference room. Some of the other user devices may have multiple microphones. If the other user device has multiple microphones, a VQS for the audio from each microphone may be determined, and a determination may be made as to which microphone or combination of microphones produces the best audio signal based on their respective VQSs.

[0111] At **720**, the method **700** includes determining whether the VQS of the audio from the other user device with the highest VQS of the other user devices is greater than the determined VQS of

the audio from the conference room device. If the VQS of the audio from the microphone of the other user device with the highest VQS of the other user devices is determined to be greater than the VQS of the audio from the microphone of the conference room device, then the method **700** includes capturing audio from the microphone of the other user device that has the highest VQS of the other devices for output to the conference at **722**. If the VQS of the audio from the microphone of the other user device that has the highest VQS of the other user devices is determined to be less than or equal to the VQS of the audio from the microphone of the conference room device, then the method **700** may include capturing audio using a microphone of the conference room device at **724**. [0112] In some examples, it may be determined that capturing audio from the identified user device and the other user device may produce the best audio signal. In these examples, the audio signals from the microphone of the identified user device and the other user device may be combined for output to the conference.

[0113] If it is determined at **706** that the identified user does not have a user device located within the conference room, the method **700** includes sampling audio using a microphones of other user devices located within the conference room at **716**. The method **700** may then continue with operations **718-724** as described above.

[0114] In some examples, it may be determined that capturing audio from the microphone of the identified user device and the microphone of the conference room device together may produce the best audio signal. In these examples, the audio signals from the microphones of the identified user device and the microphone of the conference room device may be combined for output to the conference.

[0115] FIGS. **8A-B** are illustrations of examples of UIs output for display in a conference for audio capture device selection. FIG. **8A** shows a UI **800** for a conference that includes multiple UI tiles **802** for various conference participants, a main speaker UI tile **804** for a conference participant who is actively speaking, and controls **806**. In this example, the main speaker UI tile **804** may be a single UI tile used to represent all of the video and audio of the participants in the conference room. It is noted that the single UI tile used to represent all of the video and audio of the participants in the conference room can be represented as one of the UI tiles **802**. Referring next to FIG. **8B**, in a typical conferencing system, when a user that is attending a conference in person in a conference room and joins the conference with a user device, a UI tile **808** representing the user of the user device is displayed in the conference even though the user is already represented by the UI tile for the conference room. Accordingly, the user would be represented in a single conference room UI tile, such as the main speaker UI tile **804** in this example, and the UI tile **808**. If the user has multiple user devices located within the conference room, each user device would be represented as an individual UI tile in a typical conferencing system. This can lead to unnecessary clutter in the UI output **800**. In the implementations disclosed herein, individual tiles for user devices, such as UI tile **808**, that are located within the conference room are not displayed in the conference.

Accordingly, when user devices are located within the conference room and available for audio capture for output to the conference, the UI output **800** shown in FIG. **8A** would be displayed such that user devices located within the conference room are not shown as individual UI tiles.

[0116] Disclosed herein are, inter alia, implementations of systems and techniques for device management, for example, by configuring any device such as a telephony device or a video device to operate on a virtual meeting room platform.

[0117] One aspect of this disclosure is a method for audio capture in a conference. The method includes capturing first audio using a microphone of a first device located within a conference room. The method includes determining that permission to access a microphone of a second device located in the conference room has been granted. The method includes sampling audio signals from the first device and the second device. The method includes determining a first quality score (QS) for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device. The method includes capturing second audio using the microphone of the

second device based on the first QS being less than the second QS.

[0118] Another aspect of this disclosure is a system for audio capture in a conference. The system includes a memory and a processor. The processor is configured to execute instructions stored in the memory. The processor is configured to capture first audio using a microphone of a first device located within a conference room. The processor is configured to determine that permission to access a microphone of a second device located in the conference room has been granted. The processor is configured to sample audio signals from the first device and the second device. The processor is configured to determine a first QS for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device. The processor is configured to capture second audio using the microphone of the second device based on the first QS being less than the second QS.

[0119] Another aspect of this disclosure includes a non-transitory computer-readable medium storing instructions operable to cause one or more processors to perform operations. The operations include capturing first audio using a microphone of a first device located within a conference room. The operations include determining that permission to access a microphone of a second device located in the conference room has been granted. The operations include sampling audio signals from the first device and the second device. The operations include determining a first QS for the sampled audio signals from the first device and a second QS for the audio signals from the second device. The operations include capturing second audio using the microphone of the second device based on the first QS being less than the second QS.

[0120] An aspect includes a method for audio capture in a conference comprising sampling audio signals from a first device and a second device, wherein the first device and the second device are associated with a single user. The method may include determining a first QS for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device. The method may include outputting the second audio to the conference based on the first QS being less than the second QS.

[0121] An aspect includes a system for audio capture in a conference comprising a memory and a processor. The processor may be configured to execute instructions stored in the memory to sample audio signals from a first device and a second device, wherein the first device and the second device are associated with a single user. The processor may be configured to determine a first QS for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device. The processor may be configured to output the second audio to the conference based on the first QS being less than the second QS.

[0122] An aspect may include a non-transitory computer-readable medium comprising instructions operable to cause one or more processors to perform operations. The operations may include sampling audio signals from the first device and a second device, wherein the first device and the second device are associated with a single use. The operations may include determining a first QS for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device. The operations may include outputting the second audio to a conference based on the first QS being less than the second QS.

[0123] In one or more aspects, the permission to access the microphone of the second device is granted prior to a start of the conference or subsequent to the start of the conference. One or more aspects includes comparing the first QS to a threshold and sampling the audio signals from the second device based on the first QS being below the threshold. One or more aspects includes determining that the second device is within the conference room based on calendar data associated with a user of the second device and based on a determination that the second device has moved within a predetermined time of a start time of the conference. One or more aspects includes determining that the second device is within the conference room based on calendar data associated with a user of the second device and based on geolocation data of the second device at a start time of the conference. One or more aspects includes performing keyword detection to determine a

keyword or phrase that indicates that the captured audio from the microphone of the first device is of low quality, and capturing of the second audio via the second device is based on the keyword detection. One or more aspects includes that the first audio and the second audio are captured simultaneously and combined to form a combined audio signal.

[0124] In one or more aspects, a processor is configured to compare the first QS to a threshold and sample the audio signals from the second device based on the first QS being below the threshold. In one or more aspects, the processor is configured to determine that the second device is within the conference room based on calendar data associated with a user of the second device and based on a determination that the second device has moved within a predetermined time of a start time of the conference. In one or more aspects, the processor is configured to determine that the second device is within the conference room based on calendar data associated with a user of the second device and based on geolocation data of the second device at a start time of the conference. In one or more aspects, the processor is configured to perform keyword detection to determine a keyword or phrase that indicates that the captured audio from the microphone of the first device is of low quality, wherein the capturing of the second audio via the second device is based on the keyword detection. In one or more aspects, the processor is configured to capture the first audio and the second audio simultaneously and combine the first audio and the second audio to form a combined audio signal.

[0125] In one or more aspects, one or more processors are configured to perform operations including comparing the first QS to a threshold and sampling the audio signals from the second device based on the first QS being below the threshold. In one or more aspects, one or more processors are configured to perform operations including determining that the second device is within the conference room based on calendar data associated with a user of the second device and that the second device has moved within a predetermined time of a start time of the conference. In one or more aspects, one or more processors are configured to perform operations including determining that the second device is within the conference room based on calendar data associated with a user of the second device and based on geolocation data of the second device at a start time of the conference. In one or more aspects, one or more processors are configured to perform operations including performing keyword detection to determine a keyword or phrase that indicates that the captured audio from the microphone of the first device is of low quality, wherein the capturing of the second audio via the second device is based on the keyword detection.

[0126] The implementations of this disclosure can be described in terms of functional block components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes, routines, or other programming elements.

[0127] Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc. Likewise, the terms “system” or “tool” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a

combination of software and hardware. In certain contexts, such systems or mechanisms may be understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.

[0128] Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be a device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with a processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.

[0129] Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media, and can include volatile memory or non-volatile memory that can change over time. The quality of memory or media being non-transitory refers to such memory or media storing data for some period of time or otherwise based on device power or a device power cycle. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.

[0130] While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Claims

1. A method for audio capture in a conference, the method comprising: sampling audio signals from a first device and a second device, wherein the first device and the second device are associated with a single user; determining a first quality score (QS) for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device; and outputting the second audio to the conference based on the first QS being less than the second QS.
2. The method of claim 1, wherein permission to access a microphone of the second device is granted prior to a start of the conference or subsequent to the start of the conference.
3. The method of claim 1, further comprising: determining that the second device is within a conference room based on calendar data of the single user.
4. The method of claim 1, further comprising: determining that the second device is within a conference room based on calendar data associated with the single user and based on a determination that the second device has moved within a predetermined time of a start time of the conference.
5. The method of claim 1, further comprising: determining that the second device is within a conference room based on calendar data associated with the single user and based on geolocation data of the second device at a start time of the conference.
6. The method of claim 1, further comprising: performing keyword detection to determine a keyword or phrase that indicates that the captured audio from a microphone of the first device is of low quality, wherein the capturing of the second audio via the second device is based on the keyword detection.
7. The method of claim 1, wherein the first audio and the second audio are captured simultaneously and combined to form a combined audio signal.
8. A system for audio capture in a conference, the system comprising: a memory; and a processor configured to execute instructions stored in the memory to: sample audio signals from a first device

and a second device, wherein the first device and the second device are associated with a single user; determine a first quality score (QS) for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device; and output the second audio to the conference based on the first QS being less than the second QS.

9. The system of claim 8, wherein a permission to access a microphone of the second device is granted prior to a start of the conference or subsequent to the start of the conference.

10. The system of claim 8, wherein the processor is configured to: determine that the second device is within a conference room based on location information.

11. The system of claim 8, wherein the processor is configured to determine that the second device is within a conference room based on calendar data associated with the single user and based on a determination that the second device has moved within a predetermined time of a start time of the conference.

12. The system of claim 8, wherein the processor is configured to determine that the second device is within a conference room based on calendar data associated with the single user and based on geolocation data of the second device at a start time of the conference.

13. The system of claim 8, wherein the processor is configured to: perform keyword detection to determine a keyword or phrase that indicates that the captured audio from a microphone of the first device is of low quality, wherein the capturing of the second audio via the second device is based on the keyword detection.

14. The system of claim 8, wherein the processor is configured to: capture the first audio and the second audio simultaneously; and combine the first audio and the second audio to form a combined audio signal.

15. A non-transitory computer-readable medium storing instructions operable to cause one or more processors to perform operations comprising: sampling audio signals from a first device and a second device, wherein the first device and the second device are associated with a single user; determining a first quality score (QS) for the sampled audio signals from the first device and a second QS for the sampled audio signals from the second device; and outputting the second audio to a conference based on the first QS being less than the second QS.

16. The non-transitory computer-readable medium of claim 15, wherein a permission to access a microphone of the second device is granted prior to a start of the conference or subsequent to the start of the conference.

17. The non-transitory computer-readable medium of claim 15, wherein the instructions cause the one or more processors to perform operations comprising: determining that the second device is within a conference room based on movement information.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions cause the one or more processors to perform operations comprising: determining that the second device is within a conference room based on calendar data associated with the single user and that the second device has moved within a predetermined time of a start time of the conference.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions cause the one or more processors to perform operations comprising: determining that the second device is within a conference room based on calendar data associated with the single user and based on geolocation data of the second device at a start time of the conference.

20. The non-transitory computer-readable medium of claim 15, wherein the instructions cause the one or more processors to perform operations comprising: performing keyword detection to determine a keyword or phrase that indicates that captured audio from a microphone of the first device is of low quality, wherein the capturing of the second audio via the second device is based on the keyword detection.
